Bayesian Flight Risk and Efficiency Optimiser: An AI Approach to Safer and Greener Aviation

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Problem Statement

- Aviation must balance safety with reducing environmental impact.
- Accident risk depends on factors like weather, aircraft age, and maintenance.
- Current tools separate safety and emissions analysis.
- This project uses a Bayesian AI model to estimate both accident probability and CO_2 impact.
- Supports more informed and sustainable aviation decision-making.

Introduction

- Develop a unified Bayesian AI model that estimates both flight accident probability and carbon emissions using historical flight, weather, and maintenance data.
- Continuously update predictions as new evidence is added, providing transparent, interpretable, and realistic aviation-focused risk and sustainability insights.
- Deliver a web-based dashboard where users can input flight parameters and visualise real-time risk and emissions outputs for use by airlines, analysts, and training institutions.

Aims and Objectives

Aims:

- To develop a Bayesian-based AI model for flight accident probability estimation.
- To integrate a fuel efficiency and emissions analysis component.
- To build an interactive dashboard for visualizing flight safety and environmental performance.

Objectives:

- Conduct a detailed literature review on Bayesian inference in aviation.
- Collect and preprocess aviation safety and emission datasets.
- Build a Bayesian network model for accident probability.
- Create an emissions estimation model using aircraft and route data.
- Integrate both models into one unified framework.
- Develop a Streamlit-based interactive interface.
- Validate results using real-world data.
- Evaluate ethical and professional implications in AI-driven aviation systems.

Key Literature

[1] J. Pearl, Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference, Morgan Kaufmann, 1988.
[2] ICAO, Environmental Report 2022, International Civil Aviation Organization, 2022.
[3] M. Filippone and A. Sgarro, "Bayesian Risk Modelling for Aviation Safety Analysis," Journal of Aerospace Information Systems, vol. 18, no. 2, pp. 95–107, 2021.
[4] FAA, Aviation Safety Information Analysis and Sharing (ASIAS) Program Overview, Federal Aviation Administration, 2023.
[5] Eurocontrol, Environmental Performance Report, Eurocontrol, 2020.

Methodology

- Updates predictions based on incoming evidence.
- Uses integrated aviation datasets for risk estimation.
- Refines probabilities as new data appears.
- Combines safety and emissions modelling in one framework.
- Models uncertainty through Bayesian inference.

Legal, ethical and professional

- Uses GDPR-compliant public aviation data.
- Academic-only system with transparent AI outputs.
- Mitigates bias through careful data/model checks.
- Aligns with professional and sustainability standards.

Use of AI

This project uses Bayesian inference to model uncertainty and continuously update predictions with new flight, weather, maintenance, and emissions data. The AI estimates accident probability and CO_2 output, validated against real-world datasets. Outputs are visualized with probability distributions for transparency, while human expertise remains central, ensuring AI functions as a decision-support tool, not an autonomous system.